FitBuddy-AI Fitness Plan Generator Using Gemini Models

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Project Testing Phase

Here's the *Project Testing Phase* for FitBuddy – AI Fitness Plan Generator, structured to ensure quality, reliability, and user satisfaction:

🧪 Project Testing Phase: FitBuddy

1. Testing Objectives

- Verify that all *functional requirements* are met (workout generation, nutrition tips, feedback updates, admin dashboard).

- Ensure *non-functional requirements* (performance, usability, security) are satisfied.

- Identify and fix bugs before deployment.

- Validate that the system works seamlessly across different devices and browsers.

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2. Types of Testing

- *Unit Testing*

  - Test individual functions like generate_workout_gemini(), generate_nutrition_tip_with_flash(), and update_workout_plan().

- Validate database operations (save_user(), save_plan(), update_plan()).

- *Integration Testing*

- Confirm smooth interaction between FastAPI routes, Gemini APIs, and SQLite DB.

- Example: User input → AI generation → DB storage → frontend rendering.

- *System Testing*

- End-to-end workflow validation: input form → workout plan generation → feedback update → admin view.

- Ensure all modules work together without errors.

- *User Acceptance Testing (UAT)*

- Collect feedback from real users (fitness enthusiasts, trainers).

- Check usability, clarity of plans, and responsiveness of UI.

- *Performance Testing*

- Measure response times for AI calls.

- Stress test with multiple users simultaneously.

- *Security Testing*

- Validate environment variable handling for API keys.

- Ensure user data is stored securely in SQLite.

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3. Test Scenarios

- *Scenario 1:* User submits valid details → receives a structured 7-day workout plan + nutrition tip.

- *Scenario 2:* User submits feedback → updated plan generated and stored.

- *Scenario 3:* Admin views dashboard → sees all users with original and updated plans.

- *Scenario 4:* Invalid input (e.g., missing age) → system returns error message gracefully.

- *Scenario 5:* API failure simulation → fallback error handling tested.

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4. Tools & Methods

- *Testing Tools*

  - Pytest for unit and integration tests.

  - Postman for API endpoint validation.

  - Selenium for frontend UI testing.

- *Methods*

  - Black-box testing for user-facing features.

  - White-box testing for backend logic.

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5. Deliverables

- Test cases and scripts.

- Bug reports and resolution logs.

- Final test summary report.

- UAT feedback documentation.